

# CMSE 801- Day 1

Welcome to Intro to Computational Modeling and Data Analysis!

Department of CMSE

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## The Team

- Instructor: Dr. Yann-Meing Law
- Teaching assistant: James Lennon

## Three Threads of CMSE 801

### 1. Coding in Python

By the end of the CMSE 201 you will be a proficient programmer/coder in Python. You should be able to use Python to help you in all of future classes.

### 2. Modeling

You will learn how to build models that allow you to quantitatively solve complex problems in science, engineering and beyond. Combined with the coding aspect of the course, these models will be turned into computational models.

### 3. Data Analysis

You will learn how to handle real-world data: find it, analyze it, and draw quantitative conclusions from it. You will use your coding skills for this analysis and build models that reveal deeper insights into the data.

## Course Structure

- This course will require a lot of work. There is a huge amount to learn!
- **Flipped classroom:** The class is structured to have pre-class assignments, which you will do before coming to class and in-class assignments done during class, often as part of a group.
- There is **homework every other week**.
- You will have a **midterm exam** Oct 19 but no final exam.

- Be mindful of **deadlines!**
- There is a **project**: projects will be presented at the end of the semester, although we will start planning your projects as soon as possible.
- Also, the only way to learn to code is to code. We will code in class every day. Bring your computer with you every day; we will have a few backups if you need one.

## Flipped Classroom

- The flipped classroom environment will like feel very different than what you may be used to
- Learning takes place in a collaborative manner in which you **learn by doing**, using knowledge you acquire before coming to class
- Pre-class assignments are critical for maximizing your in-class experience
- You are **responsible for monitoring your own learning** — only **you** can ensure that you are mastering the content!
- You will work in teams.
- On D2L, in the "Documents" section, there is a PDF called "Classroom Participation".

## Desire2Learn

- Will host all of the content (pre-class, in-class, and homework assignments)
- Notebooks should be downloaded from there and uploaded to Jupyter in order to run them
- **You need to drop your assignments in D2L!**

## Communication Channels

- Office hours
- Emails (inform me of your course!)
- Slacks

## Question/comments from Pre-semester Survey

- The gap compared to other students, I am transitioning to quantitative research, with this process beginning just a year ago, and so I fear that I may be slow in terms of my progress.
- That I will have a hard time keeping up since I don't have any experience with coding or computational science.
- I worry about the time needed outside of class required for course assignments while trying to continue my research projects.
- Complex theories behind specific models being difficult to comprehend / Mathematics / Whether the course might be 'too' much challenging for a social science student
- I worry that I may fall behind, and that my skills are not necessarily ready for the course
- The Professor and TA not taking enough time to make sure students (especially those without computational backgrounds) understand the course material well

- I have no background in computational methods, so I worry that I will fall behind others in the course
- I fear how will i balance this course and my research

## Getting Started

- You should have installed Anaconda on your laptops before coming to class today
- You should also have received instructions for getting this set up via email

## Day-01 In-Class Assignment: Sorting Algorithm

- Algorithm vs Computer programs
- Designing an algorithm is VERY IMPORTANT!
- Prevent bugs...